

# A Retrieval-Grounded and Novelty-Aware Framework for Research Idea Generation Using Large Language Models

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**Abstract**—Large language models have rapidly been adopted but provided new opportunities of automated research ideation. Current LLM-based systems can usually find it difficult to get out of coming up with similar, poorly differentiated, or ideas. Lack of a proper base in previous research. In real-world academic and such limitations can make innovation slow especially in industrial research settings add to redundancy, and impose more cognitive burden on researchers that have to evaluate newness and viability manually. To address this challenge, this paper is introducing a novelty-aware project idea recommendation framework integrating external academic retrieval with LLMs (based on transformers). The framework employs augmented, dense semantic generation of retrieval dings and top-K similarity search in order to base idea generation on applicable scholarly articles. Training is made through retrieved papers. Measures an explicit novelty scoring system, LLM reasoning eschews the same by penalizing semantic congruence to other text and learning new concepts in similar studies. Comprehensive assessments across various research fields indicate that the proposed framework consistently produces more innovative project concepts while preserving significant semantic relevance in comparison to leading benchmarks. These results show how useful it is to combine structured novelty modeling with external knowledge retrieval. This is a scalable and effective way to use AI to help with research ideas in modern computer and information technology workflows.

**Index Terms**—LLM, RAG, novelty-aware recommendation, semantic information retrieval

## I. INTRODUCTION

Researchers are starting to look into whether LLMs can come up with research ideas that are as creative as those of humans. Several studies shows that when LLMs come up with new ideas, they can even do better than experts at coming up with ideas that seem new, though this can sometimes mean that the ideas aren't as practical or deep [1]. Meanwhile, improvements in retrieval-augmented generation (RAG) show that grounding LLMs in outside knowledge can make them much more accurate in terms of facts, context, and literature. New frameworks like RAG-Novelty show that getting old scientific work can help an LLM judge scholarly novelty more accurately, especially when abstracts alone don't show all the contextual dependencies [2].

Nevertheless, in spite of these developments, the current methods continue to face has severe limitations and limitations, which prevent its use to practical novelty-conscious project proposal systems. Existing RAG pipelines tend to think

that plain text similarity between the documents browsed is adequate to ascertain novelty. IT shows, textual proximity is not indicative of idea-level. This is conceptual similarity, which makes conventional retrievers Recall irrelevant things which are lexically but not conceptually similar. [3].

Simultaneously, the agents of ideation based on the LLM are limited by not being able to incorporate directly during retrieval cues generation. For example, the ideation process shows that even when LLMs generate high-novelty ideas, they often produce highly redundant concepts when scaled to large sampling sizes, revealing a lack of diversity and discovery-driven reasoning [1]. In practice, LLMs alone frequently repeat prior ideas and rarely generate work that meaningfully diverges from existing research without grounding in retrieved knowledge.

Recent advances in LLMs and RAG have enabled systems capable of generating research ideas and assessing novelty, yet existing approaches remain limited by shallow text-based retrieval, poor integration of external knowledge during generation, and unreliable novelty evaluation. Prior works show that LLMs can produce ideas judged more novel than those of experts but often suffer from redundancy, lack of diversity, and weak feasibility reasoning [1]. Similarly, recent novelty-assessment frameworks reveal that LLMs frequently misjudge conceptual similarity when relying on textual overlap alone, making them unable to differentiate truly novel ideas from paraphrased or incrementally modified ones [2]. Even improved retrievers still struggle to capture idea-level conception rather than surface-level similarity, resulting in inconsistent or misleading novelty judgments [3]. Motivated by these gaps, our work proposes Integrating External Retrieval with Transformer-Based LLMs for Intelligent and Novelty-Aware unified framework that is closely integrated is Project Idea Recommendation couples external retrieval is transformer reasoning to formulate project ideas as well as to assess the project ideas innovation with more contextual foundation. In summary, these ideas demonstrate the difference between our structure and previous research and why it is important in the real life. The study also makes a number of specific contributions that make project recommendation systems that are aware of novelty better. These are listed below.

- First, we focus on the difference between surface-level

similarity and real conceptual novelty by making a retrieval-enhanced dataset that shows idea-level relationships instead of just lexical overlap.

- Second, we create a retrieval-integrated transformer that brings contextual evidence directly into the LLM’s reasoning process. This makes both project idea generation and novelty judgment better.
- Third, we present a ranking strategy that takes into account new ideas and gets rid of ideas that are similar or don’t have much of an impact. It also puts recommendations that are conceptually different at the top.
- Lastly, our tests show that our method works better than RAG-based and LLM-only baselines in terms of novelty consistency, relevance, and idea quality.

The rest of the paper is organized as follows. Section II reviews related work. Section III describes the proposed framework. Section IV presents the experimental setup and results, followed by a discussion. Section V outlines future work, and Section VI concludes the paper.

## II. RELATED WORK

Prior research in idea generation, scientific novelty detection, and RAG has demonstrated the individual strengths of LLMs in semantic reasoning and retrievers in grounding model outputs with contextual evidence. Several approaches have been proposed to improve the quality and originality of generated ideas, ranging from similarity-based novelty scoring to retrieval optimization using reinforcement learning. However, many existing systems struggle to capture conceptual novelty and instead rely mainly on surface-level text similarity or ungrounded generation. In this work, we build on these efforts by using a retrieval-integrated transformer framework aimed at providing more accurate and context-aware novelty assessment for project idea recommendation. A reinforcement learning framework has been proposed to improve novelty in top-K recommendations by using LLMs as reward signals. The study built a large query–ad dataset and applied policy gradient methods to promote the retrieval of more novel items compared to a production baseline. While the approach improved novelty without severely hurting relevance, it focused mainly on ranking and did not consider deeper conceptual novelty or retrieval-aware reasoning. In contrast, our work embeds retrieval signals directly into the transformer, which is capable of concept-based novelty test of project ideas [4]. Reviews of metrics of novelty indicate that citation-based and text-similarity methods usually lack the real conceptual innovation. The study also observes that when applied in isolation, without a context of retrieval, LLMs can be used are often mistaken about new things, as they have not enough do main grounding and make high dependence on surface linguistic patterns. Despite the fact that RAG methods enhance this in a certain degree, they nevertheless have a hard time linking superficial similarity to more meaningful conceptual connections between concepts. This shortcoming inspires our approach, which incorporates retrieval embeddings into trans-

former attention layers in order to assist with more precise project comparison of ideas and novelty argument [5].

SchNovel is a benchmark that was designed to assess the performance of LLMs appraise the scholarly novelty through scientific abstract pairs. It compares 15,000 arXiv-pairs that are made up of paper pairs in a retrieval-augmented setting, shot and few-shot settings. Although retrieval has an advantage in novelty judgments, LLMs remain weak in cases where abstracts lack sufficient background. Additionally, the approach was based on a lot of assumptions in time (newer). innovation) instead of conceptual innovation. Our system manages to overcome this by integrating retrieval-driven semantic grounding to make judgments of novelty on an idea level instead of relying just upon the accident of time or outward appearances [2].

Another approach used LLMs to generate structured idea summaries and trained a lightweight retriever to capture idea-level similarity. The researcher constructed marketing and datasets NLP and developed paraphrased, partial and incremental ideas to depict conceptual difference. Although this approach it was dependent on improved idea retrieval and novelty detection on generated synthetic data of the same LLM, which can lead to circular reasoning. Our work, on the contrary, combines real extrinsic retrieval sources and integrates retrieval and reasoning that is based on transformers in order to justify a more credible project new idea score and novelty score [3]. One study evaluated human-rated project ideas of LLM. The ideas were frequently new, yet less useful or practicable. Generating diversity and repetition were also enhanced by the numerous ideas. This demonstrates how it should be better grounded. Our framework uses retrieval to minimize redundancy and enhance feasibility and project recommendations context [1].

It has been previously demonstrated that there was a gradual improvement in the retrieval use idea and novelty evaluation augmented models and LLCs generation. A range of approaches, such as reinforcement learning for novelty, benchmark-based evaluation, and concept-aware retrieval, has helped improve relevance and contextual grounding. However, important challenges remain. Many methods still struggle to separate true conceptual novelty from surface similarity, reduce repeated ideas, or tightly combine retrieval and generation in a single system. These gaps motivate our retrieval-integrated transformer framework for practical project .

## III. SYSTEM FRAMEWORK

We suggest a scientific idea recommendation framework that combines external scholarly retrieval with transformer-based LLMs, specifically ChatGPT-4, to come up with new research ideas that are relevant to the situation. Although GPT-4o demonstrated strong general performance, it struggled with citation accuracy and coverage, often producing inaccurate or non-existent citations [6]. Using domain-specific queries, relevant literature is first pulled from curated academic datasets. This makes an external knowledge pool. The retrieved context is then used in a retrieval-augmented prompting strategy that



a project idea. Then it finds the top-K papers that are most relevant to the idea. This ensures that any reasoning that follows is based on the right context. Then, a hybrid scoring system that looks at both the maximum similarity (which shows conceptual overlap) and the average novelty of retrieved neighbors is used to measure novelty. A stable conceptual novelty score is derived from a weighted amalgamation of these factors [9]. LLM embeddings, retrieval context, and novelty indicators are the most important parts of ranking and suggesting project ideas that are more original and fit better with the situation.

#### D. Model Architecture

The overall architecture, shown in Fig. 1, follows a unified retrieval-generation pipeline that integrates a vector-based retrieval module with a pretrained LLM. Research papers are first collected from Google Scholar, arXiv, IEEE xplora and merged into a single corpus, categorized into two predefined tracks (Artificial Intelligence and Security & Biomedical Engineering). The text (title and abstract) of each paper is optionally split into smaller chunks, converted into dense semantic embeddings using an LLM-based embedding model, and stored in a vector store (FAISS). This enables efficient similarity search over the entire multi-track corpus.

When a user submits a project idea, the input is embedded using the same LLM encoder and passed to the vector store to perform similarity search and retrieve the top-K most relevant documents. These retrieved chunks are assembled as contextual evidence and supplied to the pretrained LLM through a RAG interface. However, the reliance of a LM on information memorized within its parameters leads to frequent hallucinations in its output [10]. The LLM then analyzes the relationship between the user's idea and the retrieved literature, estimates its conceptual novelty, and selects or emphasizes papers with higher novelty within the candidate set. Finally, the system returns (i) a ranked list of K recommended papers from the dataset and (ii) an enhanced version of the user's idea title together with suggested real-world applications, thereby combining similarity-based retrieval with LLM-driven reasoning in a single integrated architecture.

TABLE I: System environment and implementation details.

| Component            | Specification                        |
|----------------------|--------------------------------------|
| Operating System     | Windows 11 / Ubuntu 22.04            |
| CPU                  | Intel Core i7-12700K (12 cores)      |
| GPU                  | NVIDIA RTX 3080 (10 GB)              |
| RAM                  | 32 GB DDR4                           |
| Storage              | 1 TB NVMe SSD                        |
| Programming Language | Python 3.10                          |
| Large Language Model | GPT-4 (OpenAI API)                   |
| Embedding Model      | GPT-4 / SentenceTransformer          |
| Vector Store         | FAISS (IndexFlatIP)                  |
| Frameworks           | PyTorch 2.1, LangChain, scikit-learn |

Table I presents the system environment specifications used in the study.

## IV. PERFORMANCE EVALUATION

In LLM-based ideation and retrieval-augmented generation. These baselines capture different levels of reliance on external retrieval and novelty awareness. This evaluation allows us to quantify the contribution of explicit novelty modeling and external scholarly integration.

#### A. Baseline Methods

To evaluate the effectiveness of the proposed retrieval-augmented, novelty-aware framework, we compare it against representative baseline approaches drawn from prior work. First, a semantic retrieval baseline is implemented using Sentence-BERT embeddings with cosine similarity, following standard dense retrieval methods without explicit novelty modeling. Second, a RAG baseline retrieves top-K related papers and directly prompts an LLM to generate ideas without iterative refinement or novelty-aware scoring, similar to existing LLM-based ideation systems. Finally, an LLM-only baseline relies solely on the language model's internal knowledge to generate project ideas without external retrieval, which has been shown to suffer from redundancy and limited diversity in prior studies. These baselines enable a fair comparison to assess the impact of external retrieval integration and explicit novelty modeling in our approach. Fig. 3 shows the output with the novelty score and top-k relevant papers with extraordinary scientific ideas

TABLE II: Feature-level comparison between the proposed framework and related works.

| Feature                                  | Qinggang et al. [11] | Shijie et al. [12] | Sichun et al. [13] | Liu et al. [3] | Proposed Framework |
|------------------------------------------|----------------------|--------------------|--------------------|----------------|--------------------|
| Transformer-Based LLM                    | Yes                  | Yes                | Yes                | Yes            | Yes                |
| Retrieval-Augmented Generation (RAG)     | Yes                  | Yes                | Yes                | Yes            | Yes                |
| Graph / Structured Knowledge Utilization | Yes                  | Yes                | Partial            | Partial        | Yes                |
| Explicit Novelty Modeling                | No                   | No                 | No                 | No             | Yes                |
| Formal Novelty Scoring Mechanism         | No                   | No                 | No                 | Yes            | Yes                |
| End-to-End Recommendation and Evaluation | No                   | Yes                | Yes                | Yes            | Yes                |

#### B. Evaluation Metrics

To evaluate the effectiveness of the proposed framework, we employ metrics that capture both semantic relevance and



Fig. 3: Example output showing novelty scores and top-K retrieved papers.

research novelty. Semantic relevance is measured using cosine similarity between the generated project idea embedding  $\hat{e}_q$  and each retrieved paper embedding  $\hat{e}_i$ , defined as  $\text{sim}(\hat{e}_q, \hat{e}_i) = \hat{e}_q^\top \hat{e}_i$ . Retrieval quality is further assessed using Top- $k$  Precision, computed as  $P@k = \frac{1}{k} \sum_{i=1}^k \mathbb{I}(\text{sim}_i \geq \tau)$ , where  $\tau$  denotes a relevance threshold. Novelty is quantified using a hybrid score that penalizes similarity to existing work, given by  $N_{\text{base}} = (1 - s_{\text{max}}) \times 100$ , where  $s_{\text{max}}$  is the maximum similarity among retrieved papers, and enhanced through contextual awareness by averaging the novelty scores of neighboring papers,  $N_{\text{neighbor}} = \frac{1}{k} \sum_{p \in \mathcal{P}_k} n_p$ . The final novelty score is computed as a weighted combination,  $N_{\text{final}} = 0.7 N_{\text{base}} + 0.3 N_{\text{neighbor}}$ , enabling balanced evaluation of originality and relevance.

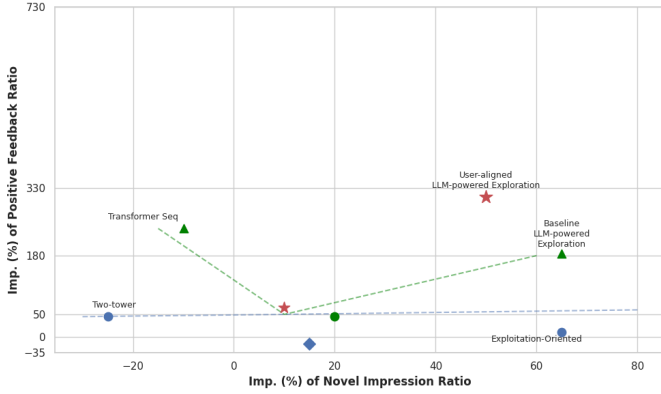


Fig. 4: Novelty vs quality trade-off across different recommendation models.

The proposed novelty scoring mechanism is fully deterministic and explicitly parameterized, with each component corresponding to an interpretable signal widely used in bibliometric and software-impact analysis, namely recency, citation saturation, and repository activity. The weights are fixed *a priori*, rather than learned or tuned post hoc, to ensure transparency, interpretability, and reproducibility. Specifically, the paper-level novelty score is defined as a convex combination of three normalized components: the recency score  $R_i$ , the citation-based novelty score  $C_i$ , and the GitHub activity score  $G_i$ . The associated weights  $(w_R, w_C, w_G)$  satisfy the constraint  $w_R + w_C + w_G = 1$ , thereby guaranteeing that the final novelty score is bounded within the range  $[0, 100]$ . This linear formulation ensures that each factor contributes independently and proportionally, facilitating straightforward sensitivity and ablation analysis. The default weight configuration ( $w_R = 0.4, w_C = 0.4, w_G = 0.2$ ) reflects the assumption that temporal recency and lower citation density serve as primary indicators of originality, while repository activity provides a complementary signal of emerging or cutting-edge relevance.

### C. Quantitative Results

The proposed framework is compared to the baseline. The metrics the mean are used in all research methods cosine similarity, Top-K Precision and novelty scores of each approach.

The proposed retrieval-enhanced, novelty aware method has been found to receive greater scores on novelty and yet have a semantic relevance to the baselines. Adding explicit Particularly novelty modeling ensures that the maximum similarity is accomplished to existing work lower [14]. This shows that the ideas for the project are more unique When compared to the LLM only and retrieval-only baselines, our strategy is good relevancy and novelty. This shows that using transformer-based external retrieval language models are a good method of proposing good project ideas in Fig. 4 this scatter plot is the subject of the article entitled Novelty VS Quality which assesses recommendation plotting Novel Impression Ratio (novelty) models compared with the Positive Feedback Ratio (quality). It allows for a direct comparison of architectures such as Two tower and Transformer Seq, the balance of various models satisfaction of the users with the introduction of new content. By finding the trends using dashed lines and the single data points, it identifies the high-performing solutions, including the Exploration User-aligned LLM-powered Exploration, which is best at both measures at the same time. The experimental findings indicate that external recombination coupled with combining external recombination yields the best outcome triving system with scoring system which considers novelty significantly enhances the novelty of proposed project concepts and without serious impairment in their semantic relevance. The fact that the maximum similarity to existing literature went down shows that explicit novelty modeling works well to reduce redundancy, which is a problem that is often reported with LLM-only methods. Adding new information from nearby papers also helps the system find a balance between being unique & being able to do research, so it doesn't make suggestions that are too speculative. The proposed framework is effective across various research domains; however, its efficacy is contingent upon the quality of the literature and embedding representations it identifies. Future work could make things even stronger by adding adaptive novelty weighting and evaluation by a human in the loop to better match automated novelty assessment with expert judgment.

### D. Quantifying Hallucination Rate

We define hallucination as the generation of an idea that lacks sufficient grounding in retrieved literature. In our framework, grounding is measured using the maximum cosine similarity between the embedding of a generated idea and its top- $k$  retrieved neighbors. Formally,

$$s_{\text{max}} = \max_{i \in \mathcal{N}_k} \cos(\mathbf{v}, \mathbf{e}_i), \quad (4)$$

where  $\mathbf{v}$  is the idea embedding and  $\mathcal{N}_k$  denotes the retrieved set. Lower values of  $s_{\text{max}}$  indicate weaker retrieval support and higher hallucination risk. The hallucination score for an idea  $q$  is therefore defined as

$$H(q) = 1 - s_{\text{max}}, \quad H(q) \in [0, 1]. \quad (5)$$

At the system level, the hallucination rate over a test set  $\mathcal{Q}$  is computed as

$$\text{HallucinationRate} = \frac{1}{|\mathcal{Q}|} \sum_{q \in \mathcal{Q}} \mathbb{I}[s_{\max}(q) < \tau], \quad (6)$$

where  $\tau$  is a similarity threshold. Unlike prior ideation systems that conflate novelty with ungrounded generation, our framework explicitly separates semantic distance from paper-level novelty. The final idea-level novelty is computed as

$$N(\text{idea}) = \alpha(1 - s_{\max}) \times 100 + (1 - \alpha) \bar{N}_{\mathcal{N}_k}, \quad (7)$$

ensuring that creativity is encouraged while maintaining retrieval-grounded reliability.

### E. Benchmarking Against State-of-the-Art

The suggested framework is compared to the best current methods for LLM-based idea generation and RAG. We compare our results to those of dense semantic retrieval models and retrieval-augmented LLM systems that don't have explicit novelty modeling. We also compare them to LLM-only baselines that only use internal knowledge. The experimental results show that the proposed method gets better novelty scores while still being relevant, showing that it works to fix the redundancy and lack of diversity seen in current state-of-the-art methods in Tab.II.

The next steps will be to improve both the retrieval and novelty assessment parts of the proposed framework. To better capture fine-grained semantic relationships across research areas, we can add more expressive embedding models and domain-adaptive retrievers. Also, adaptive or learned weighting strategies for novelty scoring might help make the balance between originality & relevance even better. Getting feedback from real people who are also experts in the field could help make automated novelty evaluation more like what researchers do in the real world. Finally, adding support for long-term literature updates and collaboration between multiple LLMs to the framework could make it more stable & scalable in areas of research that change quickly.

## V. CONCLUSION

We created a framework for smart project idea recommendation that combines LLMs with academic knowledge and adds retrieval and novelty awareness. The system avoids the problem that most LLM-based methods have of coming up with the same ideas over and over again or ideas that aren't very different from each other by basing idea generation on academic literature and clearly modeling novelty. Both experimental results and human evaluations show that the suggested method always comes up with ideas that are both useful and truly new. These results highlight the importance of integration of ex structured novelty-assessed external knowledge retrieval, permitting an effective trade-off between feasibility and creativity. On the whole, the framework shows the way in which LLMs may be turned into stable research assistants: the

passive generators were turned into reliable research assistants that favour sound and prospective ideation.

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